ALUMNICONNECT

Conceptual Entity-Relationship Diagram & Domain Model

A detailed conceptual model of the Alumni–Student Mentorship and Mock Interview Platform.

Document	Conceptual ER Diagram and Relationship Specification
Project	AlumniConnect - Alumni–Student Mentorship Platform
Version	1.0 | 02 November 2026
Model coverage	24 conceptual entities | 36 relationships | 4 domain diagrams
Primary database	MySQL 8.x with InnoDB - authoritative relational system of record

Purpose. Establish the business entities, the meaning of their relationships, and the minimum/maximum participation of each entity before converting the model into relational tables.

Prepared from the supplied “Alumni–Student Mentorship Platform: DBMS Project Blueprint”. Where the blueprint gives an entity but does not fully define a cardinality, the modelling assumption is stated explicitly in this document.

1. Model scope and reading guide

This is a **conceptual ER model**: it describes the platform domain and its business relationships, not the final physical MySQL schema. Entity identifiers and example attributes are included to make each concept understandable. Data types, indexes, SQL column definitions and implementation-specific foreign-key names are intentionally omitted.

What the model covers

Domain	Entities included	Purpose
Identity and reference	Role, User, Student Profile, Mentor Profile, Department, Company	One account identity, role-specific profiles and academic/professional reference data.
Skills and domains	Skill, Student Skill, Mentor Skill, Mentorship Domain, Student Domain Interest, Mentor Domain Expertise	Many-to-many expertise, skills and career interests.
Discovery and allocation	Mentorship Request, Recommendation Run, Recommended Mentor, Mentor Allocation	Requests, candidate recommendations, ranking history and accepted relationships.
Scheduling and sessions	Mentor Availability, Booking, Mentorship Session	Published slots, planned appointments and actual offline meeting outcomes.
Feedback and operations	Session Feedback, Mentor Review, Notification, Notification Outbox, Audit Log	Structured feedback, ratings, reliable event delivery and historical traceability.

Notation used in diagrams

Notation	Meaning
1	Exactly one related instance is required.
0..1	Optional relationship: zero or one related instance.
0..*	Optional one-to-many: zero, one or many related instances.
1..*	At least one related instance is required (not used as a default in this model).
Associative entity	An entity that resolves a many-to-many relationship and may hold relationship-specific attributes.

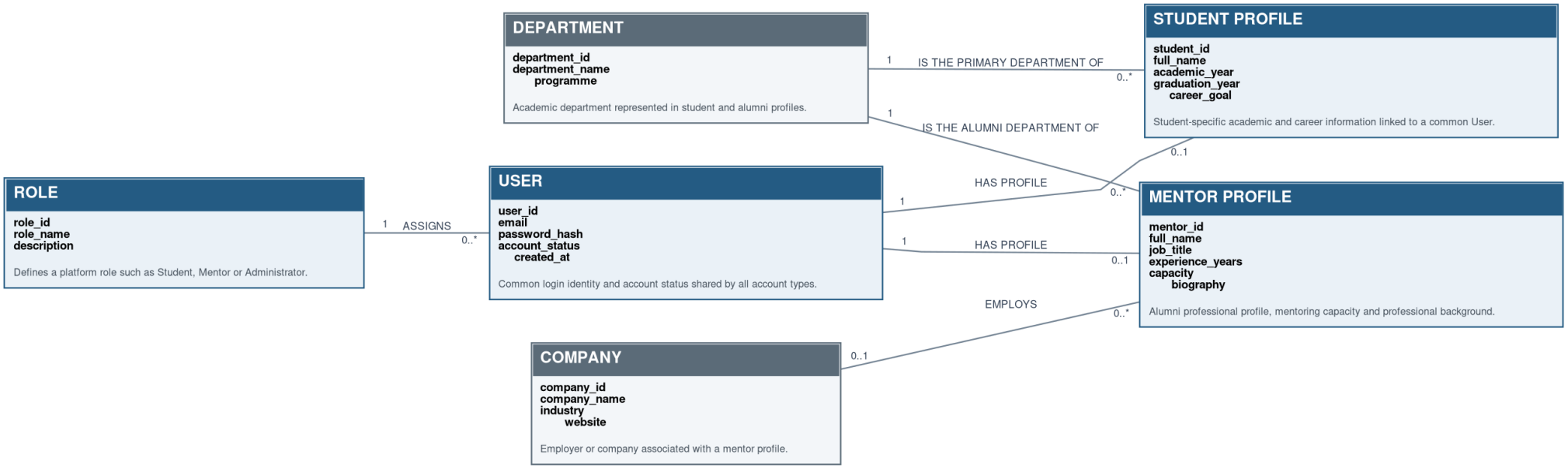
How to read a relationship: In “Department 1 — 0..* Student Profile”, one department may be associated with many student profiles, while each student profile references exactly one primary department. Read the endpoint cardinality next to the entity at that end of the line.

Important architecture boundary: MySQL owns authoritative identities, profiles, requests, allocations, bookings, sessions, feedback and reviews. Redis contains rebuildable caches and temporary state; Neo4j contains a relationship-oriented projection for recommendations. Redis and Neo4j are storage technologies, not additional business entities in this conceptual ER diagram.

Source basis: The supplied “Alumni–Student Mentorship Platform: DBMS Project Blueprint”, particularly its project vision and roles (pp. 1–3), conceptual entities and relationships (pp. 9–13), graph domain (pp. 14–19), workflows (pp. 20–27) and scope (pp. 32–34). Cardinalities not fixed by the blueprint are labelled as modelling assumptions later in this document.

Diagram 1 - Identity, profiles and reference data

One shared account identity supports role-specific student and alumni profiles, connected to department and company reference data.

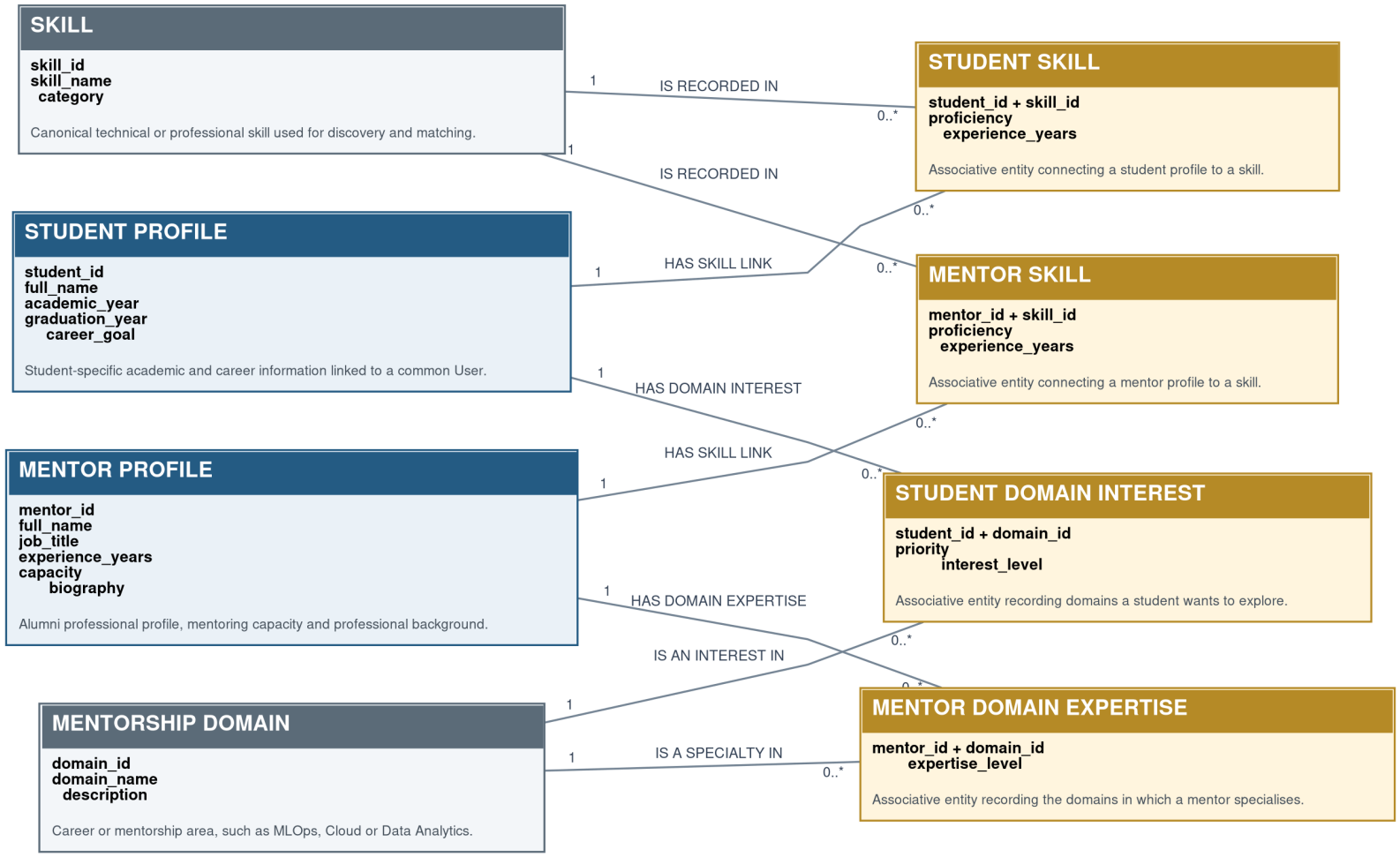


Endpoint labels show minimum..maximum participation. Entity cards show logical identifiers and example attributes; they do not define SQL data types.

Profiles and users may reappear on multiple diagram pages to keep each business domain readable; they refer to the same conceptual entities.

Diagram 2 - Skills, expertise and mentorship domains

Associative entities resolve the many-to-many relationships between students, mentors, skills and mentorship domains.

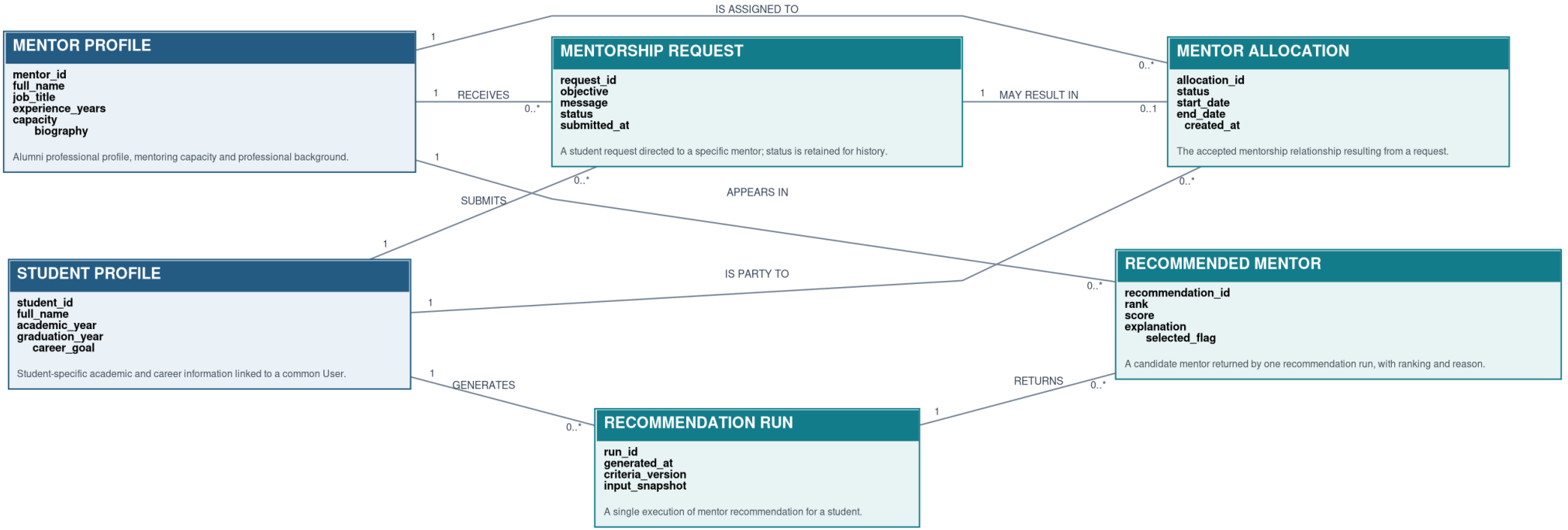


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Diagram 3 - Mentor discovery, recommendations and allocation

A request, repeated recommendation runs, candidate recommendations and the accepted allocation are separate concepts.

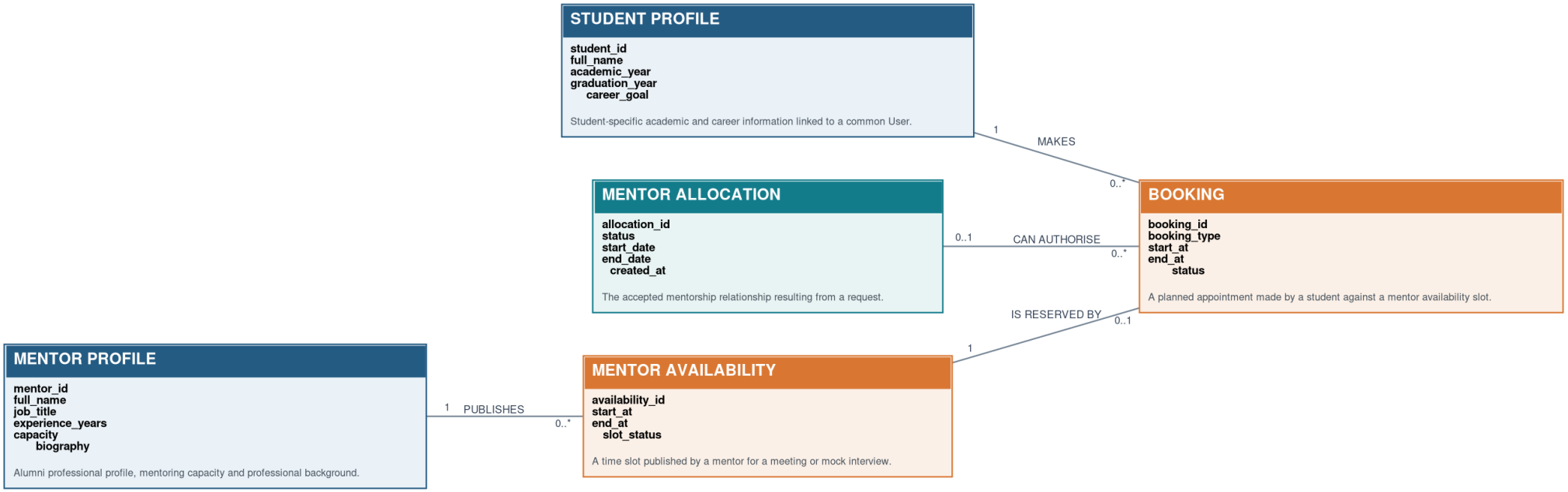


Endpoint labels show minimum..maximum participation. Entity cards show logical identifiers and example attributes; they do not define SQL data types.

Profiles and users may reappear on multiple diagram pages to keep each business domain readable; they refer to the same conceptual entities.

Diagram 4 - Mentor availability and booking

Mentors publish available slots; students reserve a slot for an appointment. MySQL must enforce one booking per slot.

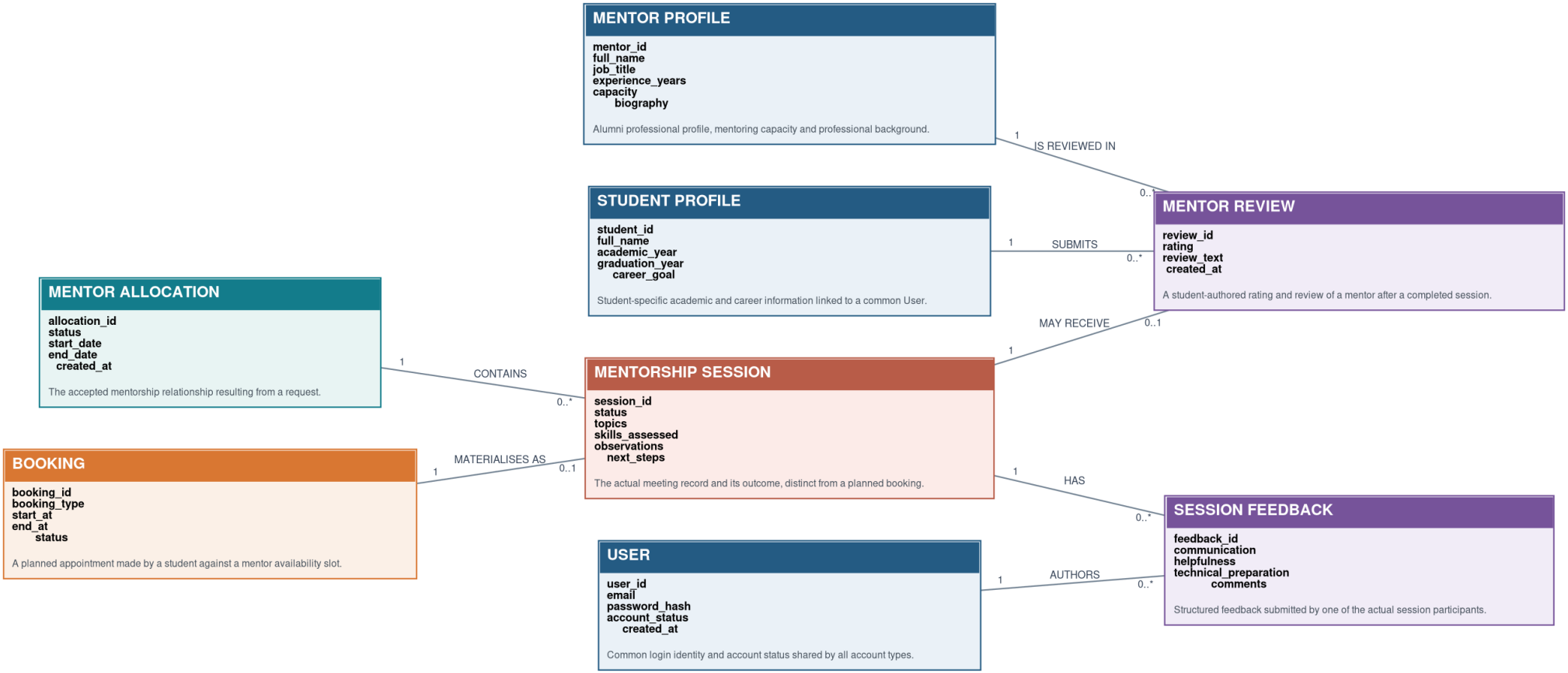


Endpoint labels show minimum..maximum participation. Entity cards show logical identifiers and example attributes; they do not define SQL data types.

Profiles and users may reappear on multiple diagram pages to keep each business domain readable; they refer to the same conceptual entities.

Diagram 5 - Sessions, feedback and mentor reviews

A session is the actual meeting record. Structured participant feedback and a student review of the mentor are modelled separately.

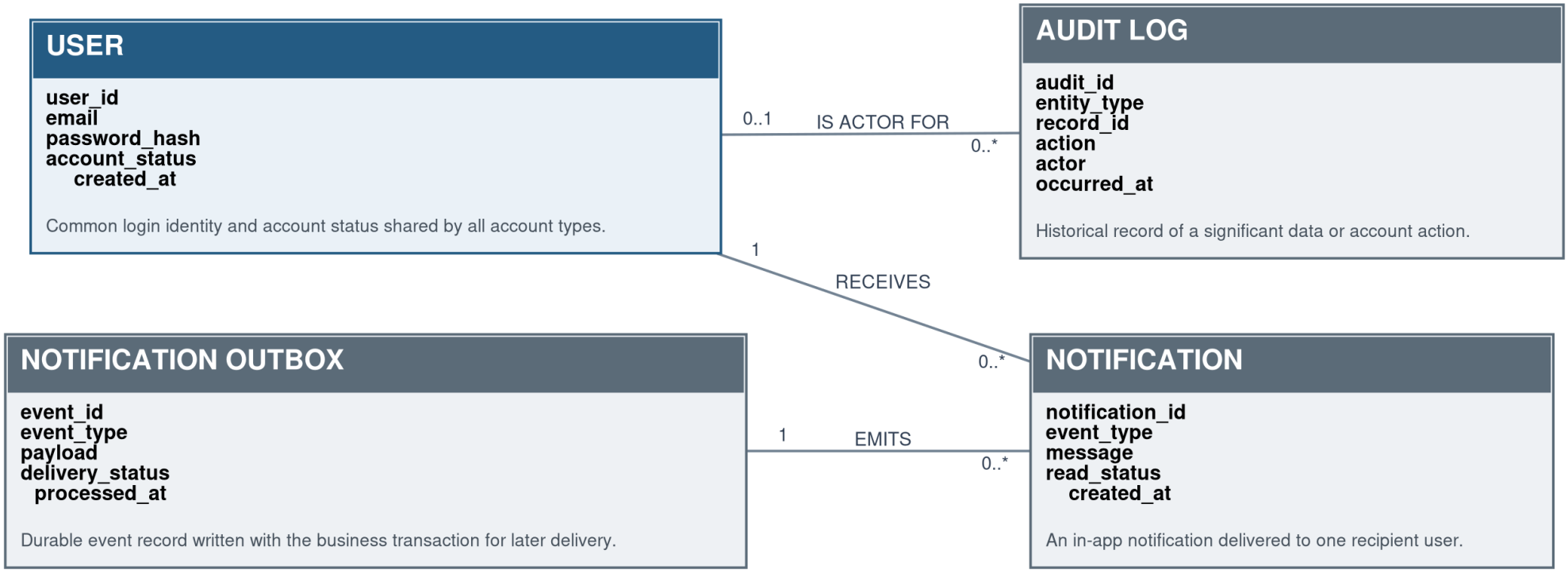


Endpoint labels show minimum..maximum participation. Entity cards show logical identifiers and example attributes; they do not define SQL data types.

Profiles and users may reappear on multiple diagram pages to keep each business domain readable; they refer to the same conceptual entities.

Diagram 6 - Notifications and audit trail

The transactional outbox records events reliably; recipient notifications and user-attributed audit records are modelled separately.



Endpoint labels show minimum..maximum participation. Entity cards show logical identifiers and example attributes; they do not define SQL data types.

Profiles and users may reappear on multiple diagram pages to keep each business domain readable; they refer to the same conceptual entities.

2. Entity catalogue - Part 1

Entity	Domain	Conceptual identifier / example attributes	Responsibility
ROLE	Identity	role_id; role_name; description	Defines a platform role such as Student, Mentor or Administrator.
USER	Identity	user_id; email; password_hash; account_status; created_at	Common login identity and account status shared by all account types.
STUDENT PROFILE	Identity	student_id; full_name; academic_year; graduation_year; career_goal	Student-specific academic and career information linked to a common User.
MENTOR PROFILE	Identity	mentor_id; full_name; job_title; experience_years; capacity; biography	Alumni professional profile, mentoring capacity and professional background.
DEPARTMENT	Reference data	department_id; department_name; programme	Academic department represented in student and alumni profiles.
COMPANY	Reference data	company_id; company_name; industry; website	Employer or company associated with a mentor profile.
SKILL	Reference data	skill_id; skill_name; category	Canonical technical or professional skill used for discovery and matching.
STUDENT SKILL	Associative	student_id + skill_id; proficiency; experience_years	Associative entity connecting a student profile to a skill.

Identity principle: User stores the shared account identity. Student Profile and Mentor Profile store role-specific information, avoiding duplicate login credentials. A user can have neither profile (for an administrator), or one of each only if the product later permits dual-role accounts.

3. Entity catalogue - Part 2

Entity	Domain	Conceptual identifier / example attributes	Responsibility
MENTOR SKILL	Associative	mentor_id + skill_id; proficiency; experience_years	Associative entity connecting a mentor profile to a skill.
MENTORSHIP DOMAIN	Reference data	domain_id; domain_name; description	Career or mentorship area, such as MLOps, Cloud or Data Analytics.
STUDENT DOMAIN INTEREST	Associative	student_id + domain_id; priority; interest_level	Associative entity recording domains a student wants to explore.
MENTOR DOMAIN EXPERTISE	Associative	mentor_id + domain_id; expertise_level	Associative entity recording the domains in which a mentor specialises.
MENTORSHIP REQUEST	Discovery	request_id; objective; message; status; submitted_at	A student request directed to a specific mentor; status is retained for history.
RECOMMENDATION RUN	Discovery	run_id; generated_at; criteria_version; input_snapshot	A single execution of mentor recommendation for a student.
RECOMMENDED MENTOR	Associative	recommendation_id; rank; score; explanation; selected_flag	A candidate mentor returned by one recommendation run, with ranking and reason.
MENTOR ALLOCATION	Discovery	allocation_id; status; start_date; end_date; created_at	The accepted mentorship relationship resulting from a request.

Recommendation principle: Recommendation Run records one execution. Recommended Mentor records each candidate returned by that run, including rank, score and explanation. This preserves recommendation history rather than overwriting the latest result.

4. Entity catalogue - Part 3

Entity	Domain	Conceptual identifier / example attributes	Responsibility
MENTOR AVAILABILITY	Scheduling	availability_id; start_at; end_at; slot_status	A time slot published by a mentor for a meeting or mock interview.
BOOKING	Scheduling	booking_id; booking_type; start_at; end_at; status	A planned appointment made by a student against a mentor availability slot.
MENTORSHIP SESSION	Session	session_id; status; topics; skills_assessed; observations; next_steps	The actual meeting record and its outcome, distinct from a planned booking.
SESSION FEEDBACK	Feedback	feedback_id; communication; helpfulness; technical_preparation; comments	Structured feedback submitted by one of the actual session participants.
MENTOR REVIEW	Feedback	review_id; rating; review_text; created_at	A student-authored rating and review of a mentor after a completed session.
NOTIFICATION	Operations	notification_id; event_type; message; read_status; created_at	An in-app notification delivered to one recipient user.
NOTIFICATION OUTBOX	Operations	event_id; event_type; payload; delivery_status; processed_at	Durable event record written with the business transaction for later delivery.
AUDIT LOG	Operations	audit_id; entity_type; record_id; action; actor; occurred_at	Historical record of a significant data or account action.

Planned vs actual: Booking represents a planned appointment. Mentorship Session represents the actual meeting and its outcome. Cancelled bookings may have no session record.

5. Relationship catalogue - Part 1

ID	From entity	From end	Relationship	To end	To entity	Interpretation / rule
R01	ROLE	1	ASSIGNS	0..*	USER	Each account has one primary role; a role can classify many users. *Assumption: one primary role per account.*
R02	USER	1	HAS PROFILE	0..1	STUDENT PROFILE	A user may have at most one student profile; each student profile belongs to one user.
R03	USER	1	HAS PROFILE	0..1	MENTOR PROFILE	A user may have at most one mentor profile; each mentor profile belongs to one user.
R04	DEPARTMENT	1	IS THE PRIMARY DEPARTMENT OF	0..*	STUDENT PROFILE	A department can be associated with many students; each student profile has one primary department. *Assumption: one primary department per student.*
R05	DEPARTMENT	1	IS THE ALUMNI DEPARTMENT OF	0..*	MENTOR PROFILE	A department can have many alumni mentors; each mentor profile records one primary alumni department. *Assumption: one primary alumni department.*
R06	COMPANY	0..1	EMPLOYS	0..*	MENTOR PROFILE	A company can be associated with many mentors; a mentor profile may have zero or one current company. *Previous employment history is not modelled.*
R07	STUDENT PROFILE	1	HAS SKILL LINK	0..*	STUDENT SKILL	A student can have multiple Student Skill association records; each record belongs to one student.
R08	SKILL	1	IS RECORDED IN	0..*	STUDENT SKILL	A skill can be linked to many students; each Student Skill record references one skill.
R09	MENTOR PROFILE	1	HAS SKILL LINK	0..*	MENTOR SKILL	A mentor can have multiple Mentor Skill association records; each record belongs to one mentor.
R10	SKILL	1	IS RECORDED IN	0..*	MENTOR SKILL	A skill can be linked to many mentors; each Mentor Skill record references one skill.
R11	STUDENT PROFILE	1	HAS DOMAIN INTEREST	0..*	STUDENT DOMAIN INTEREST	A student can express interest in multiple domains; each interest row belongs to one student.
R12	MENTORSHIP DOMAIN	1	IS AN INTEREST IN	0..*	STUDENT DOMAIN INTEREST	A domain can be selected by many students; each interest row names one domain.

Cardinality is shown from the source entity to the target entity. For example, “1” at the source and “0..*” at the target means one source record may be related to zero or many target records, while each target record refers back to exactly one source record.

6. Relationship catalogue - Part 2

ID	From entity	From end	Relationship	To end	To entity	Interpretation / rule
R13	MENTOR PROFILE	1	HAS DOMAIN EXPERTISE	0..*	MENTOR DOMAIN EXPERTISE	A mentor can specialise in multiple domains; each expertise row belongs to one mentor.
R14	MENTORSHIP DOMAIN	1	IS A SPECIALTY IN	0..*	MENTOR DOMAIN EXPERTISE	A domain can be associated with many mentors; each expertise row names one domain.
R15	STUDENT PROFILE	1	SUBMITS	0..*	MENTORSHIP REQUEST	A student may submit multiple requests; each request is submitted by one student.
R16	MENTOR PROFILE	1	RECEIVES	0..*	MENTORSHIP REQUEST	A mentor may receive many requests; each request is directed to one mentor.
R17	STUDENT PROFILE	1	GENERATES	0..*	RECOMMENDATION RUN	A student may run mentor discovery repeatedly; each run belongs to one student.
R18	RECOMMENDATION RUN	1	RETURNS	0..*	RECOMMENDED MENTOR	A recommendation run may return zero or many candidates; each candidate belongs to one run.
R19	MENTOR PROFILE	1	APPEARS IN	0..*	RECOMMENDED MENTOR	A mentor can appear in many recommendation runs; each candidate row names one mentor.
R20	MENTORSHIP REQUEST	1	MAY RESULT IN	0..1	MENTOR ALLOCATION	A request can result in at most one allocation; every allocation traces to one request. *Rejected/cancelled requests have no allocation.*
R21	STUDENT PROFILE	1	IS PARTY TO	0..*	MENTOR ALLOCATION	A student may have multiple allocations over time; each allocation identifies one student.
R22	MENTOR PROFILE	1	IS ASSIGNED TO	0..*	MENTOR ALLOCATION	A mentor may have multiple allocations subject to capacity; each allocation identifies one mentor.
R23	MENTOR PROFILE	1	PUBLISHES	0..*	MENTOR AVAILABILITY	A mentor can publish multiple slots; each availability slot belongs to one mentor.
R24	STUDENT PROFILE	1	MAKES	0..*	BOOKING	A student may make multiple bookings; each booking is made by one student.

Cardinality is shown from the source entity to the target entity. For example, “1” at the source and “0..*” at the target means one source record may be related to zero or many target records, while each target record refers back to exactly one source record.

7. Relationship catalogue - Part 3

ID	From entity	From end	Relationship	To end	To entity	Interpretation / rule
R25	MENTOR AVAILABILITY	1	IS RESERVED BY	0..1	BOOKING	A slot can be used by at most one booking; every booking reserves one slot. *Must be enforced transactionally.*
R26	MENTOR ALLOCATION	0..1	CAN AUTHORISE	0..*	BOOKING	An allocation may authorise multiple appointments; a booking may be linked to zero or one allocation. *Booking-to-allocation is optional in this conceptual model when the appointment is otherwise authorised.*
R27	MENTOR ALLOCATION	1	CONTAINS	0..*	MENTORSHIP SESSION	An allocation may have multiple actual sessions; each session belongs to one allocation.
R28	BOOKING	1	MATERIALISES AS	0..1	MENTORSHIP SESSION	A confirmed booking may produce one session record; each session comes from one booking. *Cancelled bookings need not have a session.*
R29	MENTORSHIP SESSION	1	HAS	0..*	SESSION FEEDBACK	A session may receive multiple structured feedback entries; each entry concerns one session.
R30	USER	1	AUTHORS	0..*	SESSION FEEDBACK	A user may author multiple feedback entries; each entry has one author. *Author must be an actual session participant.*
R31	MENTORSHIP SESSION	1	MAY RECEIVE	0..1	MENTOR REVIEW	A session may have at most one student review; each review refers to one completed session. *Based on one student participant per session.*
R32	STUDENT PROFILE	1	SUBMITS	0..*	MENTOR REVIEW	A student may submit multiple reviews across sessions; each review has one student author.
R33	MENTOR PROFILE	1	IS REVIEWED IN	0..*	MENTOR REVIEW	A mentor may receive many reviews; each review evaluates one mentor.
R34	USER	1	RECEIVES	0..*	NOTIFICATION	A user may receive many notifications; each notification has one recipient.
R35	NOTIFICATION OUTBOX	1	EMITS	0..*	NOTIFICATION	One outbox event can fan out to multiple recipient notifications; each notification traces to one event.
R36	USER	0..1	IS ACTOR FOR	0..*	AUDIT LOG	A user may generate many audit entries; an audit entry may have no user actor when system-generated.

Cardinality is shown from the source entity to the target entity. For example, “1” at the source and “0..*” at the target means one source record may be related to zero or many target records, while each target record refers back to exactly one source record.

8. Business rules that constrain the ER model

These rules turn the conceptual relationships into behaviours the application must preserve. They should later become database constraints, transaction rules, API validations and test cases where appropriate.

Rule	Requirement
BR-01 - Account identity	Each account has one common User identity. Student and mentor information belongs in separate profile entities, not duplicated login records.
BR-02 - Role separation	The model assigns one primary role to each user: Student, Mentor or Administrator. The blueprint does not specify multi-role accounts; a UserRole associative entity would be needed if multiple roles per account are required.
BR-03 - Profile cardinality	A User may have zero or one Student Profile and zero or one Mentor Profile. An Administrator can therefore exist without either profile.
BR-04 - Skills and domains	Student Skill, Mentor Skill, Student Domain Interest and Mentor Domain Expertise each resolve a many-to-many association. A profile-skill or profile-domain pair should not be duplicated.
BR-05 - Recommendation history	A Recommendation Run may return many Recommended Mentor candidates. Candidate rank, score and explanation are preserved per run so later runs do not erase prior results.
BR-06 - Request lifecycle	A Mentorship Request progresses through Pending, Accepted, Rejected or Cancelled. Rejected and cancelled requests remain available for history and analytics.
BR-07 - Allocation creation	A request results in at most one Mentor Allocation. Acceptance verifies that the request is still pending, checks mentor capacity, prevents duplicate active allocations and commits request status plus allocation atomically.
BR-08 - Availability ownership	Every Mentor Availability slot belongs to one Mentor Profile. A slot can be booked at most once.
BR-09 - Booking correctness	Before confirming a booking, validate the slot is still available, the student has no overlapping appointment, mentor capacity is respected, the relevant request/allocation is valid, and the submission is not a duplicate.
BR-10 - Transactional booking	Booking confirmation must be atomic. Concurrent attempts for the same slot must not both succeed. MySQL/InnoDB is authoritative; Redis cache must never decide availability.
BR-11 - Booking vs session	A booking is a plan. A session is the actual offline meeting. A confirmed booking may produce at most one session; a cancelled booking need not produce one.
BR-12 - Session lifecycle	A session can be Scheduled, Completed, Cancelled or No-show. The system preserves session history and mentor-entered topics, assessed skills, performance observations, improvement areas and next steps.
BR-13 - Feedback eligibility	Feedback cannot be submitted before a session has taken place. Only the actual student or mentor participants may author session feedback.
BR-14 - Mentor review uniqueness	A student can submit at most one mentor review per completed session. The review references the session, student author and mentor being reviewed.
BR-15 - Notification events	Requests, decisions, booking confirmations/cancellations, reminders and feedback events can produce notifications for the appropriate participants.
BR-16 - Reliable notification delivery	Write an outbox event in the same MySQL transaction as the business change. A background worker later delivers it; this avoids sending a notification for a rolled-back operation.
BR-17 - Audit history	Important status and data changes should be auditable. An audit record may have a user actor or be system-generated.
BR-18 - Source of truth	MySQL is authoritative. Redis stores temporary/rebuildable values. Neo4j is a recommendation projection and may be briefly stale; booking and allocation eligibility must be revalidated in MySQL.

9. Key lifecycle views

The following state paths are business-level illustrations from the blueprint. They are not extra entities.

Mentorship request

Initial state	Allowed outcome	ER-model consequence
Pending	Accepted	Create one Mentor Allocation; retain accepted request.
Pending	Rejected	Retain request; no allocation is created.
Pending	Cancelled	Retain request; no allocation is created.

Booking and session

Object	Illustrative states	Meaning
Booking	Pending / Confirmed / Cancelled	The scheduled appointment and reservation of a mentor availability slot.
Mentorship Session	Scheduled → Completed / Cancelled / No-show	The actual meeting record, created for a confirmed appointment and maintained as history.

Feedback and review

Object	Who can create it?	When?	Cardinality limit
Session Feedback	Either actual participant (student or mentor)	After the session has taken place	A session may have multiple feedback entries.
Mentor Review	Student participant	After a completed session	At most one student review per session.

Example traversal: a Student Profile generates a Recommendation Run. The run returns several Recommended Mentor candidates. The student submits a Mentorship Request to one mentor. If accepted, the request produces one Mentor Allocation. The mentor publishes Mentor Availability; the student makes a Booking; the confirmed booking produces a Mentorship Session. Session Feedback and a Mentor Review are then attached to the completed session.

10. Explicit modelling assumptions and open decisions

The blueprint defines the business domain and several cardinalities, but it does not specify every participation constraint. The following choices make the ERD implementable without claiming that they were all explicitly fixed in the blueprint.

ID / decision	Assumption or open decision
A1 - Primary role	One primary Role is assigned to each User. If users must act as both student and mentor under one account, introduce UserRole (M:N) or formalise a dual-profile rule.
A2 - Primary department	Each Student Profile and Mentor Profile has one primary Department. If multiple degrees/departments must be represented, introduce StudentDepartment and MentorDepartment associative entities.
A3 - Current company	Mentor Profile has zero or one current Company. Prior employment history is outside this model; add MentorEmployment if needed.
A4 - Booking authorisation	Booking may be associated with zero or one Mentor Allocation in this conceptual model. The implementation must define whether a valid accepted request alone can authorise a one-off booking.
A5 - Availability slot	Every Booking reserves exactly one availability slot; a slot can be reserved by at most one Booking. Slot duration and time-zone rules are implementation details.
A6 - One student per session	The current model assumes one student and one mentor per booking/session. Group mentoring or panel interviews would require participant association entities.
A7 - One review per session	The one-review maximum assumes a single student participant in each session. If group sessions are later supported, uniqueness must be per student-session pair.
A8 - Notification fan-out	One outbox event can produce multiple notifications, for example one to the student and one to the mentor. Each notification has exactly one recipient.
A9 - System audit actor	Audit Log actor is optional so system-generated events can be represented without inventing a human user.
A10 - Semantic vector search	Vector embeddings and vector indexes are an optional Phase 2 capability and are not included as entities in this core conceptual ER model.

11. Database responsibility and conceptual boundary

Technology	Role in this model	What it does not own
MySQL 8.x + InnoDB	Authoritative entities and business relationships: accounts, profiles, requests, allocations, availability, bookings, sessions, feedback, reviews, notifications and audit data.	No split-brain source of truth for core business state.
Redis	Rebuildable cache for mentor cards, reference data and recommendation results; temporary login/rate-limit/idempotency state.	Must not decide whether a slot is truly available or whether an allocation is valid.
Neo4j	Graph projection of Student, Mentor, Skill, Domain, Company and Department relationships for discovery and explainable recommendations.	Does not replace MySQL identity, booking or allocation authority.
Redis Stack vector search (optional)	Embeddings and semantic retrieval for free-text career goals and profile descriptions.	Phase 2 extension; not required for this core ERD.

12. Coverage and implementation hand-off

This model gives the team a stable conceptual starting point for the next DBMS artefacts: an ER diagram review, relational schema mapping, functional-dependency analysis, normalisation, SQL DDL, constraints, indexes, transaction workflows and test data. The blueprint recommends a modular monolith: business modules may be separated in code, while the primary relational data remains in one MySQL database.

Recommended next modelling step: Convert the entities and associative entities into a relational schema, define primary and foreign keys, then verify the proposed 3NF/BCNF decomposition against actual functional dependencies. Do not infer that the example attributes in this document are the final schema without that analysis.

Prepared as a conceptual design deliverable for the DBMS project. This document is a design specification, not evidence that the application or database has already been implemented.